

BIOLOGY MCQS

1. **The Science of growing terrestrial plants in an aerated solution is_____?**

A. Hydrodynamics

B. Hydroponics

C. Hydrotonics

D. None of them

2. **An educated guess is called:**

A. Accumulated Data

B. Observation

C. Hypothesis

D. Conclusion

3. **Dolly, the sheep was a highly successful clone from a somatic cell, reported back in_____?**

A. 1993

B. 1994

C. 1995

D. 1996

4. **Crop rotation helps in_____?**

A. Growth of pathogens

B. Soil Infertility

C. Slowing of crops growth

D. Eradication of disease

5. **People living in high altitudes (like mountains) usually have a _____?**

A. smaller number of Red Blood Cells

B. larger number of Red Blood Cells

C. smaller number of White Blood Cells

D. larger number of White Blood Cells

6. **Which character differentiates living things from non-living organisms?**

A. They live in the same ecosystem.

B. They are highly organized and complex made of one or more cells and contain genetic material

C. They are acted upon by the same environment

D. Both a and b

7. **Which Vitamin is needed for the formation of a light-sensitive pigment in the retina.**

A. **Vitamin A**

B. Vitamin C

C. Vitamin D

D. B complex

8. **The average internal temperature of human body is _____?**

- A. 35 oC
- B. 36 oC
- C. 37 oC**
- D. 38 oC

9. **The study of fossils is called?**

- A. Environmental Biology
- B. Historical biology
- C. Palaeontology**
- D. Social biology

10. **About _____ of the body weight of a mammal is water ?**

- A. 60%
- B. 65%
- C. 70%**
- D. 75%

11. **Tibia is a bone found in the _____?**

- A. Skull
- B. Arm
- C. Leg**
- D. Face

12. **Which is the largest gland in the human body?**

- A. Thyroid
- B. Liver**
- C. Pancreas
- D. None of these

13. **Water has maximum density at _____?**

- A. 0°C
- B. 4°C**
- C. 4° K
- D. 4.8°C

14. **The scientific study of livings is called?**

- A. Anatomy
- B. Biology**
- C. Geology
- D. Zoology

15. **The word “BIOLOGY” has been derived from _____ word?**

- A. Latin
- B. English
- C. Greek**
- D. French

16. **According to modern system of taxonomy, living organisms have been classified into?**

- A. Three Kingdoms
- B. Five Kingdoms**

- C. Two Kingdoms
- D. Six Kingdoms

17. **Kingdom Protocista includes _____?**

- A. Fungi
- B. Prokaryotes
- E. protists**
- D. Plantae

18. **Kingdom Fungi includes _____?**

- A. A cellular, eukaryotic organisms
- B. Non-chlorophylls, multicellular eukaryotic organisms
- C. Non-chlorophylls, multicellular, saprophytic organisms**
- D. Chlorophyllous, multicellular, eukaryotic Organisms

19. **Earthworm is included in _____?**

- A. Kingdom Plantae
- B. Kingdom Protocista
- C. Kingdom Monera
- D. Kingdom Animalia**

20. **The Study of organisms inhabiting the sea and Ocean and the physical and chemical characteristics of their environment is:**

- A. Social Biology
- B. Marine Biology**
- C. Environmental Biology
- D. Fresh Water Biology

21. **The branch of Biology which deals with the use of data and techniques of engineering & technology concerning living organisms is:**

- A. Human Biology
- B. Environmental Biology
- C. Genetics
- D. Bio-technology**

22. **The branch in which organism's life cycle, mode of transmission and interaction with their hosts are studied is:**

- A. Anatomy
- B. Physiology
- C. Parasitology**
- D. Social Biology

23. **The molecules of living matter that provide building blocks are mostly based on:**

- A. Carbon**
- B. Nitrogen
- C. Hydrogen
- D. Ammonia

24. **Human blood is an example of _____?**

- A. Organelle
- B. Tissues**

- C. Organic compound
- D. Inorganic compounds

25. **With different chemical arrangements and formation Of complex molecules, the life emerges on the level of the ?**

- A. **Cells**
- B. Tissues
- C. Organs
- D. Organ System

26. **Organelles are _____?**

- A. Unicellular structures
- B. Multi-cellular structures
- C. **Sub-cellular structures**
- D. Non-cellular structures

27. **Functional unit of DNA Is _____?**

- A. Nucleus
- B. **Genes**
- C. Chromosomes
- D. Nucleolus

28. **Various tissue types combine to make up _____?**

- A. Cells
- B. **Organs**
- C. Organ System
- D. Organelles

29. **Two or more populations of different species living and interacting in the same area form**

- A. Biosphere
- B. **Community**
- C. Population System
- D. Ecosystem
- E. Species

30. **A group of similar cells that perform a specific function is:**

- A. Organ
- B. **Tissues**
- C. Organelle
- D. Organ System

31. **Which of the following is an example of “molecule”?**

- A. Chloroplast
- B. A Nerve Cell
- C. **DNA**
- D. Hydrogen

32. **A group of very similar inter breeding organisms constitutes?**

- A. Biosphere
- B. Ecosystem

C. Population

D. Species

33. **Members of one species inhabiting the same are form_____?**

A. Species

B. Community

C. Ecosystem

D. Population

34. **On the basis of collected facts and figures, a scientist formulates a factitive statement called:**

A. Observation

B. Research work

C. Hypothesis

D. Inductive Reasoning

35. **Robert Hooke discovered_____?**

A. Nucleus

B. Mitochondria

C. Cell

D. DNA

36. **Steps Involved in scientific method are ?**

A. Hypothesis → Accumulated data → observation & experimentation → conclusion

B. New data → Conclusion → Accumulated data → Observation & experimentation

C. Accumulated data – New data – Observation & experimentation – Hypothesis → Conclusion

D. Accumulated data → Hypothesis → Observation & experimentation → New data → Conclusion

37. **Malaria is associated with_____?**

A. Culex Mosquito

B. Culiesta Mosquito

C. Anopheles Mosquito

D. Culex & Anopheles Mosquito

38. **Biologist ‘Grass’ discovered the life cycle of plasmodium in_____?**

A. 1896

B. 1897

C. 1898

D. 1899

39. **The first antibiotic to be discovered was_____?**

A. **Penicillin**

B. Cephalosporin

C. Amino glycosides

D. Erythromycin

40. **Immunization by vaccination was first introduced by?**

A. Fleming in 1945

B. Florey & Chain in 1975

- C. Flore & Chain in 1795
- D. Edward Jenner in 1795**

41. **The experiments on DNA molecules in chromosomes for knowing the basis of inherited diseases are conducted by ?**

- A. Molecular biologists**
- B. Microbiologists
- C. Freshwater biologists
- D. Social biologist

42. **On wound _____ fight foreign particles, like bacteria?**

- A. White Blood cells**
- B. Red Blood Cells
- C. Platelets
- D. Plasma

43. **Sythetic insulin from pork was formed by which technique?**

- A. Biotechnology**
- B. Social biological techniques
- C. Parasitology
- D. Both a and c

44. **Blood is carried towards the heart by _____?**

- A. Arteries
- B. Capillaries
- C. Veins**
- D. none of these

45. **The branch of biology which deals with the study of social behavior and communal life of human beings living in any environment is called ?**

- A. Environmental biology
- B. Social biology**
- C. Human biology
- D. Both b and c

46. **_____ is essential for the formation of hemoglobin.**

- A. Calcium
- B. Iron**
- C. Water
- D. Carbohydrates

47. **Out of 92 naturally occurring chemical elements how many are considered as bio-elements?**

- A. 6
- B. 16**
- C. 26
- D. 15

48. **_____ are needed as a source of energy for the vital activities of the body?**

- A. Calcium
- B. Iron

C. Water

D. Carbohydrates

49. **Oxygen accounts for _____ % of totals human body mass?**

A. **65%**

B. 18%

C. 10%

D. 3%

50. **The atoms of different elements combine with each other through ionic or covalent bonding to produce compounds this stable form is called:**

A. An organ

B. A molecule

C. Tissue

D. Both a and c

51. **Which one of them is a micro-molecule?**

A. Polysaccharide

B. Protein

C. Hemoglobin

D. ATP

52. **A structure formed by groups of similar cells organized into loose sheets or bundles performing similar functions is called as:**

A. An organ

B. An organism

C. A Tissue

D. A Cell

53. **Haemodialysis means cleaning of _____?**

A. Urine

B. Blood

C. Glomerular filtrate

D. Coelomic fluid

54. **In animals coordination is achieved by means of?**

A. Nervous system

B. Endocrine system

C. Respiratory system

D. Both A and B

55. **Production of sweat and sebum is related with ?**

A. **Skin**

B. Liver

C. Lungs

D. GIT

56. **Group of living organisms of the same species living in the same place at the same time is called ?**

A. Community

B. Population

- C. Ecosystem
- D. Biome

57. **The evaporative cooling in the respiratory tract of dogs is called?**

- A. Vasodilation
- B. Vasoconstriction
- C. Panting**
- D. All of these

58. **Different species of plants and animals living in the same habitat is called?**

- A. Population
- B. Community**
- C. Biome
- D. Habitat

59. **Which of the following pathogen type cause disease that can be treated with antibiotics ?**

- A. bacteria**
- B. fungi
- C. virus
- D. none of these

60. **A large regional community primarily determined by climate is called ?**

- A. Population
- B. Community
- C. Biome**
- D. Habitat

61. **Most cell membranes are composed principally of ?**

- A. DNA and protein
- B. Protein and Lipids**
- C. Protein and Chitin
- D. Protein and RNA

62. **The concept that various organisms dominated this planet during various geological time period and thus placing organisms in a time sequence came from the studies by ?**

- A. Environmental biologist
- B. Paleontologist**
- C. Marine biologist
- D. Social biologist

63. **Normally, in the process of osmosis, the net flow of water molecules into or out of the cell depends upon differences in the ?**

- A. Concentration of water molecules inside and outside the cell**
- B. Concentration of enzymes on either side of the cell membrane
- C. Rate of molecular motion on either side of the cell membrane
- D. None of these

64. **It is possible to date the rocks by comparing the amount of specific radioactive isotopes they contain. Which of the statement is correct in this respect?**

- A. Older sediment layers have equal amount of these radioactive isotopes as that of the young ones
- B. Older sediment layers have less amount of these radioactive isotopes as that of the young ones.**
- C. Older sediment layers have greater amount of these radioactive isotopes as that of the young ones.
- D. Older sediment layers did not have these radioactive isotopes

65. Sodium ions are “pumped” from a region of lower concentration to a region of higher concentration in the nerve cells of humans. This process is an example of

- A. Diffusion
- B. Passive transport
- C. Osmosis
- D. Active transport**

66. An unbroken series of species arranged in ancestors to descendent sequence with each later species having evolved from one that immediately preceded it is called?

- A. Biome
- B. Phyletic lineage**
- C. Community
- D. Population

67. Which are the four most abundant elements in living cells ?

- A. carbon, oxygen, nitrogen, sulfur
- B. carbon, oxygen, hydrogen, nitrogen**
- C. carbon, oxygen, sulfur, phosphorus
- D. carbon, sulfur, hydrogen, magnesium

68. Which one is not correct for a productive theory?

- A. It is predictive
- B. It has explanatory power
- C. It discourages suggestion of different hypotheses**
- D. None of these

69. Conclusion of Mendels work latter became a _____?

- A. Scientific hypothesis
- B. Theory
- C. Scientific law**
- D. Productive theory

70. Starch is converted into maltose by?

- A. diastase**
- B. invertase
- C. maltase
- D. amylase

71. Breeders have developed new and better varieties of food items by using which technique?

- A. Pasteurization
- B. Hydroponic culture technique
- C. Genetic engineering**
- D. Biological control techniques

72. **Co-enzyme is often formed from ?**

- A. lipid
- B. protein
- C. inorganic ion
- D. vitamin**

73. **Production of genetically identical copies of organisms/cells by asexual reproduction is called ?**

- A. Mitosis
- B. Replication
- C. Cloning**
- D. Biological control

74. **Messenger RNA is formed in_____?**

- 75. A.nucleus**
- B.chloroplast
- C.mitochondria
- D.none of these

76. **Astronauts may use which technique to grow fruits and vegetables ?**

- A. Tissue culture techniques
- B. Cloning
- C. Pasterisation
- D. Hydroponic culture technique**

77. **Number of chromosomes in E.coli ?**

- A. 1**
- B. 2
- C. 4
- D. 6

78. **Proteins are made from amino acids by the process of _____?**

- A. Hydrolysis
- B. Pinocytosis
- C. Dehydration synthesis**
- D. Active transport

79. **Biological sciences have a set methodology and it is based on ?**

- A. Experimental inquiry**
- B. Esthetic preference
- C. Philosophical ideas
- D. Imaginations

80. **Which is an organic compound found in most cells ?**

- A. Water
- B. Glucose**

- C. Oxygen
- D. Sodium chloride

81. **A series of hypothesis supported by the results of many tests is called ?**

- A. Scientific law
- B. Theory**
- C. Data
- D. Deduction

82. **Which statement is incorrect for pasteurization?**

- A. It is used to preserve yogurt and milk
- B. It was developed by Louis Pasteur.
- C. It involves heating the substance at high temperature for just few sec.
- D. It can be used to preserve vegetables and meat.**

83. **Protein factory is _____?**

- A. nucleus
- B. ribosome**
- C. golgi complex
- D. centriole

84. **What is the mode of transmission of Hepatitis virus?**

- A. Aerosol
- B. Parental (via blood)**
- C. Skin penetration
- D. Unknown

85. **Smallest disease causing agents in plants are ?**

- A. virion
- B. mycoplasma
- C. viroids**
- D. prions

86. **Which disease can be controlled by vaccination?**

- A. Measles**
- B. Cancer
- C. Diabetes
- D. Heart attack

87. **The major cell infected by the HIV is lymphocyte ?**

- A. helper-T**
- B. B
- C. both T and B
- D. none of these

88. **Vaccine was first developed by _____?**

- A. Louis Pasture
- B. Edward Jenner**
- C. Jaber Ibn Hayan
- D. Aristotle

89. **Pigment present in red algae is _____?**

- A. fucoxanthin
- B. phycocyanin
- C. phycoerythrin**
- D. bilirubin

90. **Which disease has been totally eradicated from the world because of effective vaccination?**

- A. Measles
- B. Polio
- C. Small pox**
- D. Hepatitis

91. **Nutrition in fungi is _____?**

- A. Photosynthetic
- B. Chemosynthetic
- C. Completely parasitic
- D. Absorptive heterotrophs**

92. **Which of the following statement is incorrect about antibiotics?**

- A. They are used against microorganisms
- B. They are always effective against bacteria
- C. They are used in diseases like tuberculosis and pneumonia
- D. They are derived from bacteria only**

93. **Fungi resemble plants because they lack _____?**

- A. Cell wall
- B. Cytoplasm
- C. Centriole**
- D. Nucleus

94. **Which treatment is instituted in a cancerous patient?**

- A. Antibiotic therapy
- B. Radiotherapy**
- C. Gene therapy
- D. None of these

95. **Which is a parasitic plant?**

- A. cuscuta**
- B. rose
- C. ferns
- D. mosses

96. **Which statement is incorrect for cloning?**

- A. The nucleus of a fertilized egg is replaced by the nucleus from the cell of a fully developed individual
- B. Division of a single egg into one or more separate embryos.
- C. The individual is the mirror image of the parent organism
- D. It involve methods of sexual reproduction**

97. **True roots absent in _____?**

- A. ferns
- B. bryophytes**
- C. gymnosperms
- D. angiosperms

98. **Removal or degradation of environmental pollutants or toxic materials by living organisms is called_____?**

- A. Integrated disease management
- B. Hydroponic culture technique
- C. Pasteurization
- D. Bioremediation**

99. **The mechanism for ATP synthesis is_____?**

- A. Chemosynthesis
- B. Photosynthesis
- C. Phosphorylation
- D. Chemiosmosis**

100. **Which of the following is not a viral disease?**

- A. Cowpox
- B. Mumps
- C. Tetanus**
- D. Small pox

101. **Enzyme present in the saliva is_____?**

- A. lipase
- B. trypsin
- C. ptyalin**
- D. invertase

102. **In human body 99% of total mass is formed of_____?**

- A. 16 Bio-elements
- B. 12 Bio-elements
- C. 10 Bio-elements
- D. 6 Bio-elements**

103. **Nitrogen is present in_____?**

- A. carbohydrates
- B. proteins**
- C. lipids
- D. carbonates

104. **Biological organization is_____?**

- A. simple
- B. advance
- C. complex
- D. highly complex**

105. **Operculum is present in_____?**

- A. bony fish**
- B. sea fish

- C. cartilaginous fish
- D. none of these

106. **The arrangement of _____ speaks of the division of labour within cell?**

- A. Molecule
- B. Subatomic particles
- C. Atoms
- D. Organelles**

107. **In which animal respiratory surface are found in more than one organ?**

- A. birds
- B. human
- C. fish
- D. frog**

108. **Communities are _____ collection of organisms?**

- A. Static
- B. Dynamic**
- C. May be static or dynamic
- D. None of these

109. **The heart is enclosed in a membrane called _____?**

- A. pleura
- B. pericardium**
- C. peritoneum
- D. epithelium

110. **The number of species of organisms currently known to science is?**

- A. 3500000
- B. 35000
- C. 2500000**
- D. 25000000

111. **From right ventricle blood is pushed into?**

- A. pulmonary trunk**
- B. brain
- C. aorta
- D. body

112. **What is the branch of biology in which structure and function of tissues are studied _____?**

- A. Cell Biology
- B. Taxonomy
- C. Histology**
- D. Morphology

113. **Biometry is the application of _____ in biology.**

- A. Algebra
- B. Statistics**
- C. Matrix
- D. Geometry

114. **Study of insects is called_____?**

- A. Immunology
- B. Parasitology
- C. Entomology**
- D. Paleontology

115. **The term biology was introduced in_____ in 1800?**

- A. Germany**
- B. Turkey
- C. Italy
- D. Japan

116. **Which of the following is the basic unit of classification_____?**

- A. Species**
- B. Genus
- C. Family
- D. Order

117. **Euglena belongs to the kingdom_____?**

- A. Fungi
- B. Monera
- C. Protista**
- D. Non of them

118. **Viruses are considered living organisms because_____?**

- A. They can be crystallized
- B. They occur in plants and animals
- C. They are microscopic
- D. They contain DNA and can be reproduced**

119. **All of the following are elements that plants need in very small amounts except_____?**

- A. Iron
- B. Hydrogen**
- C. Chlorine
- D. Copper

120. **Capillaries are the example of_____?**

- A. Organ System**
- B. Cell
- C. Organ
- D. Tissue

121. **Who discovered the living cell first time?**

- A. Robert brown
- B. Robert hooke**
- C. Rudolf virchow
- D. Antoni van leeuwenhoek

122. **Amoeba can move with the help of?**

- A. Flagella
- B. Pseudopodia**
- C. Water
- D. Cilia

123. **Who is Called the Father of Demography?**

- A. ALAN Turing
- B. John Dalton
- C. Charles Babbage
- D. IBN KHALDUN**

124. **The one which is present in all living things**

- A. Cell nucleus
- B. DNA or RNA**
- C. Cell membrane
- D. Vacuole

125. **The study of nature is called_____?**

- A. Science**
- B. Atom
- C. Molecule
- D. Element

126. **Animals that eat plants are called_____?**

- A. Herbivores**
- B. Consumer
- C. Carnivores
- D. Omnivores

127. _____ **contains the green chemical called chlorophyll.**

- A. Cytoplasm
- B. Cell membrane
- C. Chloroplasts**
- D. Vacuole

128. _____ **is a scientific word for human beings?**

- A. Animals
- B. Homo sepions**
- C. Omnivore
- D. Consumer

129. **Lactometer is used for_____?**

- A. Ship
- B. Milk**
- C. Honey
- D. None

130. **enzyme is_____?**

- 131. A.bio catalyst**
- B.hormone

- c. Gland
- D. None of these

132. **Exact replica of chromosome refers to _____?**

- A. **Chromatid**
- B. Centromere
- C. Kinetochore
- D. Arms

133. **Density gradient centrifugation separates the materials on the basis of _____?**

- A. Size alone
- B. Density alone
- C. **Both a & b**
- D. Velocity of movement

134. **Ribosomes are tiny Granules first studied in 1955 by _____?**

- A. De-Duve
- B. Schwann
- C. August Weismann
- D. **Palade**

135. **Virus consist of Parts: _____?**

- A. 1
- B. **2**
- C. 3
- D. 4

136. **I can break down fatty acid to succinate _____?**

- A. Peroxisome
- B. **Glyoxisome**
- C. Lysosome
- D. Golgi bodies

137. **Mitosis is divided into _____?**

- A. Karyokinesis
- B. Cytokinesis
- C. Interphase
- D. **Both a & b**

138. **Ribosomes and DNA are also present in _____?**

- A. **mitochondria**
- B. golgi complex
- C. endoplasmic reticulum
- D. lysosomes

139. **Taking in liquid material into the cells is known as _____?**

- A. Phagocytosis
- B. **Pinocytosis**
- C. Exocytosis
- D. Both a and b

140. Under a compound microscope chromosome is made of arms and_____?

- A. Centromere
- B. Centrosome
- C. Chromatid
- D. Centromere**

141. The main types of plastids are_____?

- A. Chloroplasts
- B. Chromoplasts
- C. Leucoplasts
- D. All of these**

142. Fibers of the extracellular matrix is the characteristics of_____?

- A. Plant cell wall
- B. Bacterial cell wall
- C. Cell membrane of bacteria
- D. Fluid mosaic model of cell membrane**

143. The process involved in inflammation of liver_____?

- A. apoptosis
- B. Necrosis**
- C. Both a & b
- D. Non disjunction

144. Polysome is formed during _____ process?

- A. Transcription
- B. Secretion
- C. Translation**
- D. Division

145. In 1831, the presence of nucleus in the cells was reported by_____?

- A. Robert Koch
- B. Robert Hooke
- C. Robert Mug Abe
- D. Robert Brown**

146. 7 Secretions are products formed within cell on_____?

- A. E.R
- B. Golgi bodies**
- C. Ribosome
- D. Nucleus

147. On average there are _____ or more thylakoid per granum?

- A. 60
- B. 50**
- C. 40
- D. 70

148. Un-separated replica of chromosome is called_____?

- A. Chromatin
- B. Kinetochore
- C. Chromatid**
- D. Centromere

149. **A-biogenesis means that_____?**

- A. Living things originated from bacteria
- B. Mycobionts
- C. Non-living things**
- D. None of these

150. **The tumours which are of small size and localized are_____?**

- A. **benign**
- B. malignant
- C. gentle
- D. nasty

151. **PCR stand for _____?**

- A. **Polymerase chain reaction**
- B. Poly carbomyl reaction
- C. Peptide chain reaction
- D. Pentapeptide chain reaction

152. **The innate tendency of offspring to resemble their parents is called_____?**

- A. **heridity**
- B. heridisk
- C. varaition
- D. none

153. **Ribosomes were discovered by_____?**

- A. Golgi
- B. De Duve
- C. Palade**
- D. R.Brown

154. **Which character differentiates living things from non-living organisms_____?**

- A. They live in the same ecosystem.
- B. They are acted upon by the same environment
- C. They are highly organized and complex made of one or more cells and contain genetic material**
- D. Both a and b

155. **The study of fossils is called_____?**

- A. Environmental Biology
- B. Historical biology
- C. Palaeontology**
- D. Social biology

156. **A chemical substance that reacts with enzyme but is not transformed into product and thus blocking active site is called_____?**

- A. Substrate
- B. Co-factor
- C. Inhibitor**
- D. Promotor

157. The experiments on DNA molecules in chromosomes for knowing the basis of inherited diseases are conducted by_____?

- A. Molecular biologists**
- B. Microbiologists
- C. Freshwater biologists
- D. Social biologist

158. Percentage of water in brain cells is_____?

- A. 20%
- B. 65%
- C. 85%**
- D. 89%

159. The atoms of different elements combine with each other through ionic or covalent bonding to produce compounds this stable form is called_____?

- A. An organ
- B. A molecule**
- C. Tissue
- D. Both a and c

160. Different species of plants and animals living in the same habitat is called _____?

- A. Population
- B. Community**
- C. Biome
- D. Habitat

161. In animals coordination is achieved by means of_____?

- A. Respiratory system
- B. Nervous system
- C. Endocrine system
- D. Both b and c**

162. _____ is the potential source of chemical energy for cellular activities?

- A. C-H bond**
- B. C-O bond
- C. C-N bond
- D. P-O-C bond

163. The branch of biology which deals with the study of social behavior and communal life of human beings living in any environment is called_____?

- A. Environmental biology
- B. Social biology**
- C. Human biology
- D. Both b and c

164. The concept that various organisms dominated this planet during various geological time period and thus placing organisms in a time sequence came from the studies by_____?

A. Environmental biologist

B. Paleontologist

C. Marine biologist

D. Social biologist

165. Group of living organisms of the same species living in the same place at the same time is called_____?

A. Community

B. Population

C. Ecosystem

D. Biome

166. Control of organisms by using living organisms is called_____?

A. Bioremediation

B. Integrated disease management

C. Antisepsis

D. Biological control

167. Which one is a micromolecule_____?

A. Polysaccharide

B. Protein

C. Hemoglobin

D. ATP

168. Unbroken series of organisms arranged from ancestor to descendant sequence is_____?

A. Biodiversity

B. Phyletic lineage

C. Connecting link

D. Evolutionary line

169. Out of 92 naturally occurring chemical elements how many are considered as bio-elements_____?

A. 6

B. 16

C. 26

D. 15

170. A large regional community primarily determined by climate is called_____?

A. Population

B. Community

C. Biome

D. Habitat

171. Branch of biology which deals with the study of chemical components and chemical processes in living organisms is called_____?

- A. Molecular biology
- B. Physiology
- C. Biochemistry**
- D. Atomic biology

172. **A structure formed by groups of similar cells organized into loose sheets or bundles performing similar functions is called as _____?**

- A. An organ
- B. An organism
- C. A Tissue**
- D. A Cell

173. **Oxygen accounts for _____ percent of total human body mass?**

- A. 65%**
- B. 18%
- C. 10%
- D. 3%

174. **Out of total organisms on earth, _____ are vascular plants?**

- A. 53.10%
- B. 19.90%
- C. 17.60%**
- D. 9.40%

175. **Breakdown of large molecules into smaller ones utilizing water molecules is _____?**

- A. Hydration
- B. Hydrolysis**
- C. Dehydration synthesis
- D. Electrolysis

176. **Intake of liquid material by cell membrane is called _____?**

- A. Endocytosis
- B. Exocytosis
- C. Phagocytosis
- D. Pinocytosis**

177. **Which one is not correct for a productive theory _____?**

- A. It is predictive
- B. It has explanatory power
- C. It discourages suggestion of different hypotheses**
- D. None of these

178. **_____ is concerned with cell secretions?**

- A. Endoplasmic reticulum
- B. Golgi complex**
- C. Lysosomes
- D. Cell membrane

179. **Synthetic insulin from pork was formed by which technique _____?**

- A. Parasitology
- B. Social biological techniques
- C. Biotechnology**
- D. Both a and c

180. **Cytoskeleton involved in assembly and disassembly of the spindle during mitosis is of_____?**

- A. Microtubules**
- B. Microfilaments
- C. Intermediate filaments
- D. None of these

181. _____ is used to relieve one kind of headache, migraine?

- A. Pencillin
- B. Lovastatin
- C. Cyclosporine
- D. Ergotin**

182. **Bacteria which grow either in the presence or absence of oxygen_____?**

- A. Aerobic bacteria
- B. Anaerobic bacteria
- C. Facultative bacteria**
- D. Microaerophilic

183. **It is possible to date the rocks by comparing the amount of specific radioactive isotopes they contain. Which of the statement is correct in this respect:**

- A. Older sediment layers have equal amount of these reaioactive isotopes as that of the young ones
- B. Older sediment layers have less amount of these reaioactive isotopes as that of the young ones.**
- C. Older sediment layers have greater amount of these radioactive isotopes as that of the young ones.
- D. Older sediment layers did not have these radioactive isotopes

184. **Reverse transcriptase is present in_____?**

- A. All RNA viruses
- B. All DNA viruses
- C. All retroviruses**
- D. All viruses

185. **Destruction of all life forms is called_____?**

- A. Sterilization**
- B. Disinfection
- C. Antisepsis
- D. Preservation

186. **Pox viruses are_____?**

- A. RNA non-enveloped
- B. RNA enveloped

C. DNA non-enveloped

D. DNA enveloped

187. **Pollen grains in pinus have _____ for dispersal through wind?**

A. Flagella

B. Cilia

C. Wings

D. Feathers

188. **The single healthy megaspore retained within the megasporangium germinates to form an egg containing female gametophyte called _____?**

A. Ovule

B. Seed

C. Fruit

D. Embryo sac

189. **Alternation of generations is absent in _____?**

A. Hydrant

B. Obelia

C. Jelly fish

D. Physalia

190. **Double fertilization is feature of _____?**

A. Ferns

B. Whisk ferns

C. Gymnosperms

D. Angiosperms

191. **If bile pigments are prevented from leaving digestive tract, then they cause _____?**

A. Hepatitis

B. Jaundice

C. Gall stones

D. Blockage

192. **Initial pH of food vacuole during digestion in amoeba is _____?**

A. 5.3

B. 5.6

C. 7.3

D. 7.6

193. **The arrangement of _____ speaks of the division of labour within cell?**

A. Molecule

B. Subatomic particles

C. Atoms

D. Organelles

194. **NADH is oxidized by _____?**

A. **Coenzyme ?**

B. Cytochrome b

- C. Oxygen
- D. Glycolysis

195. **Salivary glands present in front of ear are_____?**

- A. Sublingual
- B. Submaxillary
- C. Submandibular
- D. Parotid**

196. **How many spiracles are present in cockroach_____?**

- A. 5
- B. 10
- C. 20**
- D. 40

197. **Which of the following is the correct sequence in the biological method?**

- A. Observation?hypothesis?law?theory
- B. Observation?hypothesis?deduction?testing of deduction**
- C. Hypothesis?observation?deduction?testing of deduction
- D. law?theory?deduction?observation

198. **Amphibians are considered to be evolved from_____?**

- A. Newts
- B. Varanope
- C. Dipnoi**
- D. Hagfishes

199. **Enterobius vermicularis is commonly known as_____?**

- A. Flatworm
- B. Pinworm**
- C. Hook worm
- D. Tapeworm

200. **Red to orange pigments are_____?**

- A. **Carotenes**
- B. Carotenoids
- C. Xanthophylls
- D. Chlorophyll

201. **Animals of which class of arthropoda are present everywhere_____?**

- A. Crustacean
- B. Insecta**
- C. Arachnida
- D. Myriapoda

202. **Gaseous exchange in birds occurs at level of_____?**

- A. Alveoli
- B. Air sacs
- C. Bronchi
- D. Parabronchi**

203. _____ of earth surface is covered with water?

- A. 10%
- B. 30%
- C. 75%**
- D. 90%

204. **An antibody is made of _____ polypeptide chains?**

- A. 1
- B. 2
- C. 3
- D. 4**

205. **Production of genetically identical copies of organisms/cells by asexual reproduction is called _____?**

- A. Mitosis
- B. Replication
- C. Cloning**
- D. Biological control

206. **95% of the cytoplasm of RBCs is _____?**

- A. Hemoglobin**
- B. Enzymes
- C. Salts
- D. Other proteins

207. _____ cells of phloem are directly involved in transport of **organic solutes?**

- A. Parenchyma
- B. Companion cell
- C. Tracheid
- D. Sieve element**

208. **An infectious disorder of respiratory system is _____?**

- A. Cancer
- B. Asthma
- C. Tuberculosis**
- D. Emphysema

209. **Cuticular transpiration is _____ of total transpiration?**

- A. 1-2%
- B. 3-4%
- C. 5-7%**
- D. 90%

210. **According to cohesion-tension theory, tension is created by _____?**

- A. Hydrogen bonding
- B. Transpiration**
- C. Adhesion
- D. Root prerssure

211. **Deoxygenated blood first enters in _____ part in fishes?**

- A. **Sinus venosus**
- B. Right atrium
- C. Left atrium
- D. Conus arteriosus

212. **Ozone depletion occurs commonly due to _____?**

- A. Carbon dioxide
- B. **Chlorofluorocarbons**
- C. UV rays
- D. Deforestation

213. **Who coined the term CELL?**

- A. Schwann
- B. Schleiden
- C. **Robert Hook**
- D. Both a and b

214. **“lubb” sound is produced due to _____?**

- A. Opening of inlet valve
- B. **Closure of inlet valve**
- C. Opening of outlet valve
- D. Closure of outlet valve

215. **In condition of high temperature, following does not occur or occurs at low level _____?**

- A. Vasodilation
- B. Activation of sweat glands
- C. **Thermogenesis**
- D. Evaporative cooling

216. **Who first observed and thus hypothesized that new cells are formed from previously existing living cell?**

- A. Theodor Schwann and Schleiden
- B. **Rudolph Virchows**
- C. Louis Pasteur
- D. Both b and c

217. **In Hardy-Weinberg formula P² represents frequency of _____?**

- A. Recessive allele
- B. Dominant allele
- C. **Homozygous dominant individual**
- D. Homozygous recessive individual

218. **_____ supported theory of special creation?**

- A. **Linnaeus**
- B. Aristotle
- C. Mendel
- D. Lamarck

219. **Percentage of proteins in cell membrane is _____?**

- A. 20-40%
- B. 40-50%
- C. 60-80%**
- D. 80-100%

220. Which of the following is most constant abiotic component of ecosystem_____?

- A. Air
- B. Water
- C. Land
- D. Gravity**

221. Ecosystem not found in Pakistan is_____?

- A. Tropical rain forest**
- B. Temperate deciduous forest
- C. Grassland
- D. Tundra

222. Movement of the material across the cell membrane which does not requiring expenditure of metabolic energy is called_____?

- A. Active transport
- B. Passive transport**
- C. Co-transport
- D. Counter transport

223. The productivity of aquatic ecosystem is basically determined by light and_____?

- A. CO₂
- B. O₂
- C. Water
- D. Nutrients**

224. The first layer of cell wall which is formed is_____?

- A. Primary wall**
- B. Secondary wall
- C. Middle lamella
- D. All of these

225. Annual rainfall in grassland is about_____?

- A. More than 2500 mm
- B. 750-1500 mm
- C. 250-750 mm**
- D. Less than 250 mm

226. Conversion of ammonia or ammonium ion into nitrites during nitrogen cycle occurs due_____?

- A. Rhizobium
- B. Nitrosomonas**
- C. Nitrobacter
- D. Pseudomonas

227. **Cellulose is the major component of_____?**

- A. **Primary wall**
- B. Secondary wall
- C. Middle lamella
- D. all of the above

228. **Humming bird is an example of_____?**

- A. Cold blooded animal
- B. Ectotherm
- C. Endotherm
- D. **Heterotherm**

229. **Number of cervical vertebrae is_____?**

- A. 5
- B. **7**
- C. 9
- D. 12

230. **Autophagosomes are_____?**

- A. Those lysosomes which eat parts of their own cells to generate energy.
- B. Those lysosomes which eat old and worn out cellular organelles.
- C. Lysosomes which help in extracellular digestion
- D. **Both a and b**

231. **Sapwood is formed from_____?**

- A. Primary xylem
- B. **Secondary xylem**
- C. Primary phloem
- D. Secondary phloem

232. **Which of the following cytoskeletal fiber contain tubulin protein?**

- A. **One which help in assembly of spindles during mitosis.**
- B. One involved in internal cell motion.
- C. One involved in maintenance of cell shape.
- D. Both b and c

233. **Only myosine is present in_____?**

- A. A band
- B. 1 band
- C. **H zone**
- D. Sarcomere

234. **Joints present in skull are example of_____?**

- A. **Fibrous joints**
- B. Cartilaginous joints
- C. Synovial Joints
- D. Slightly moveable joints

235. **The human naked eye can differentiate between two points which are _____ apart?**

- A. **1.0 mm**
- B. 0.1 mm
- C. 1.0 cm
- D. 1.0 dm

236. _____ is a nervous disorder characterized by involuntary tremors, diminished motor power and rigidity?

- A. Epilepsy
- B. Meningitis
- C. Alzheimer's disease
- D. Parkinson's disease**

237. Of the following which one is not the characteristic of mitochondria?

- A. It contains F1 particles
- B. It is involved in the synthesis of protein
- C. It is a self replicating organelle
- D. Number of mitochondria is constant**

238. In human female, fertilization commonly occurs at _____?

- A. Proximal part of oviduct**
- B. Distal part of oviduct
- C. Uterus
- D. Cervix

239. The part of chloroplast where CO₂ is fixed to manufacture sugar is _____?

- A. Stroma**
- B. Grana
- C. Thylakoid
- D. Outer membrane

240. Highest form of learning is _____?

- A. Latent learning
- B. Insight learning**
- C. Operant conditioning
- D. Instinct

241. Which of the following hormone is polypeptide in nature _____?

- A. Insulin
- B. Thyroxine
- C. ADH**
- D. Cortisone

242. Ribosomes are assembled in _____?

- A. Nucleolus**
- B. Nucleus
- C. DNA
- D. RNA

243. Far-red light promotes flowering in _____?

- A. **Short day plant**
- B. Long day plant
- C. Day neutral plant
- D. None of these

244. **Which of the following is not present in mitochondria?**

- A. Enzymes
- B. Co-enzymes
- C. Ribosomes
- D. **Thylakoids**

245. **Secretin inhibits production of _____?**

- A. **Gastric juice**
- B. Pancreatic juice
- C. Bile
- D. All of these

246. **Yellowish glandular structure which produces progesterone is _____?**

- A. Graffian follicle
- B. Ovary
- C. **Corpus luteum**
- D. Placenta

247. **The stent energy in the form of ADP is regenerated by mitochondria into _____?**

- A. AMP
- B. ADP
- C. **ATP**
- D. All of these

248. **XO-XX pattern of sex determination is present in _____?**

- A. **Grasshopper**
- B. Butterfly
- C. Drosophila
- D. Man

249. **On which of the following component of chloroplast chlorophyll is arranged?**

- A. Cell membrane
- B. Matrix
- C. **Thylakoids**
- D. Stroma

250. **An example of codominance is _____?**

- A. A' blood group
- B. B' blood group
- C. O' blood group
- D. **AB' blood group**

251. **Test cross is used to find _____?**

- A. Trait
- B. Phenotype
- C. Genotype**
- D. Ratios

252. Number of nuclear pores/nucleus in an RBC are _____?

- A. 1?3
- B. 3?4**
- C. 10000
- D. 30000

253. Yellow cytoplasm in ascidian gives rise to _____?

- A. Epidermis
- B. Muscle cells**
- C. Gut
- D. Notochord

254. Chlorophyll molecule contrains _____ as central metal ion.

- A. Fe^{2+}
- B. Mg^{2+}**
- C. Zn^{2+}
- D. Cu^{2+}

255. Okazaki fragments are connected together through action of _____?

- A. Primase
- B. Polymerase
- C. Helicase
- D. Ligase**

256. Which of the following combination is an example of self replicating organelles?

- A. Mitochondria Ribosomes
- B. Mitochondria Nucleus
- C. Mitochondria Chloroplast**
- D. Mitochondria Vacuole

257. Which of the following is an autosomal disorder _____?

- A. Down's syndrome**
- B. Klinifelter's syndrome
- C. Turner's syndrome
- D. Jacob's syndrome

258. The place of centromere where spindle fibres get attached is _____?

- A. Kinetochore**
- B. Kinochore
- C. Centromere region
- D. all of these

259. Embryonic induction is caused by part developing from _____?

- A. Endoderm
- B. Ectoderm
- C. Mesoderm**
- D. Extraembryonic layers

260. **Sickle cell anemia is an example of _____?**

- A. Chromosomal aberration
- B. Insertion of gene
- C. Deletion of gene
- D. Point mutation**

261. **The type of plastids which help in pollination is _____?**

- A. Chromoplasts**
- B. Leucoplasts
- C. Chloroplasts
- D. All of these

262. **Purines and pyrimidines are in equal ration in DNA. It was indicated by _____?**

- A. F.Miescher
- B. Chargaff**
- C. Watson & Crick
- D. Harshay & Chase

263. **Chlorophyll is a/an _____ molecule?**

- A. Inorganic
- B. Cationic
- C. Anionic
- D. Organic**

264. **Ribonucleo-protein particles are the name of _____?**

- A. DNA
- B. Nucleus
- C. Eukaryotic ribosomes**
- D. RNA

265. **Lock and Key model was proposed by _____?**

- A. Emil Fischer**
- B. Koshland
- C. Robin Williams
- D. Rudolph Virchow

266. **Which step causes activation of catalytic site of an enzyme _____?**

- A. Change in pH of the surroundings.
- B. Formation of Enzyme Susstrate complex.**
- C. Change in the charge of the active site.
- D. Change in temperature

267. **Amoeboid movements and movement of cyclosis is due to _____?**

- A. **Microfilaments**
- B. Microtubules
- C. Intermediate filaments
- D. Cytoskeleton

268. **Enzymes** _____ the activation energy of a chemical reaction?

- A. Increases
- B. **Decreases**
- C. Does not effect
- D. Increases or decreases depending upon individual enzyme

269. **Which one forms the raw material for coenzymes**_____?

- A. **Vitamins**
- B. Carbohydrates
- C. Proteins
- D. Metals

270. **Proteins and lipids are converted into glycolipids and glycoproteins by adding carbohydrates by**_____?

- A. Ribosomes
- B. Cytoplasm
- C. **Golgi apparatus**
- D. Endoplasmic reticulum

271. **Biological molecules (proteins) which catalyze a biochemical reaction and remain unchanged after completion of reaction are called**_____?

- A. Cofactor
- B. Coenzymes
- C. Activator
- D. **Enzymes**

272. **In which of the following location enzymes controlling cellular respiration are present**_____?

- A. Nucleus
- B. Chloroplast
- C. **Mitochondria**
- D. Ribosome

273. **In golgi apparatus the maturing face is**_____?

- A. Biconcave
- B. Convex
- C. Spherical
- D. **Concave**

274. **A cofactor made of inorganic ion which is detachable is called**_____?

- A. Prosthetic group
- B. Coenzyme
- C. **Activator**
- D. Cofactor

275. An activated enzyme consisting of polypeptide chain and a cofactor is called_____?

- A. Apoenzyme
- B. Holoenzyme
- C. Activated enzyme
- D. Both b and c**

276. Centrioles are composed of _____ triplets of microtubules.

- A. 3
- B. 9**
- C. 10
- D. 15

277. Which statement about enzyme is incorrect_____?

- A. Some of them consist solely of protein with no non protein part.
- B. They catalyze a chemical reaction without being utilized.
- C. All enzymes are fibrous Proteins.**
- D. They without their cofactor are called apoenzyme.

278. A three dimensional dcavity bearing a specific charge by which the enzyme reacts with its substrate is called_____?

- A. Active site**
- B. Binding site
- C. Catalytic site
- D. Allosteric site

279. Cellular organelles related with H₂O₂ are_____?

- A. Glyoxisomes
- B. Lysosomes
- C. Peroxisomes**
- D. Ribosomes

280. If more substrate to already occurring enzymatic reaction is added more enzyme activity is seen because_____?

- A. There is probably more substrate present than there is enzyme.
- B. There is probably more enzyme available than there is substrate.**
- C. There is probably more product present than there is either substrate or enzyme.
- D. The enzyme substrate complex is probably failing to form during the reaction

281. A group of ribosomes attached to mRNA is known as_____?

- A. Polymer
- B. Polypeptide
- C. Polysomes**
- D. Monomer

282. The rate of reaction is directly proportional to the concentration of an enzyme which statement is incorrect in this respect_____?

- A. Increase in enzyme molecule increases the available active sites.
- B. This relation is for unlimited time period with unlimited enzyme concentration**
- C. If the concentration is doubled the rate will become two fold.
- D. None of these.

283. Which statement is incorrect about Lock and Key Model_____?

- A. Specific enzyme can transform only a specific substrate
- B. Active site of an enzyme is a non flexible structure
- C. Active site does not change before during or even after the reaction
- D. It explains the mechanism of every chemical reaction**

284. If the concentration of enzyme is kept constant and amount of substrate is increased a point is reached where increase in substrates concentration does not affect the reaction rate because of_____?

- A. Enzymes get denatured at higher substrate conc
- B. Rate of reaction is indirectly proportional to substrate concentration at this point
- C. All the active sites on enzyme molecule are occupied**
- D. None of these

285. The active site of an enzyme_____?

- A. Never changes
- B. Forms no chemical bond with substrate
- C. Determined by structure and the specificity of the enzyme**
- D. They are non specific in their action

286. Factory of Ribosomal sythesis is_____?

- A. Cytoplasm
- B. Nucleus
- C. Nucleolus**
- D. Endoplasmic reticulum

287. If more substrate to already occurring enzymatic reaction is added and there is no effect on the rate of the reaction what is the form given to this situation_____?

- A. Saturation**
- B. Denaturation
- C. Composition
- D. Inhibition

288. Who opposed the idea the cell is an empty space bounded by thick wall?

- A. Lorenz Oken
- B. Schwann
- C. Robert Brown**
- D. Rudolph Virchow

289. Extreme change in pH results in_____?

- A. Change in ionization of amino acids at the active site of the enzyme
- B. Change in the ionization of the substrate
- C. Denaturation of the enzyme**
- D. Increase in the reaction rate

290. Resolution power of a compound microscope is_____?

- A. 2.0 ?m**
- B. 2-4 A

- C. 24 ?m
- D. 24 A

291. **Excessive increase in temperature of medium causes the enzyme molecule to _____?**

- A. Activate
- B. Unaffected
- C. Denatured**
- D. None of these

292. **If enzyme concentration is low than substrate pH and temperature values are equal to requirement then which of the following will increase rate of reaction _____?**

- A. increase in concentration of enzyme**
- B. increase in concentration of substrate
- C. increase in pH
- D. increase in temperature

293. **In cell fractionation various components of cells including its organelles can be isolated in different layers depending upon _____?**

- A. Their physical properties like size & weight.
- B. Physical properties of the medium like its density.
- C. Their electrical properties like their charges
- D. Both a and b**

294. **The structure of an enzyme is altered by _____?**

- A. Irreversible inhibitor**
- B. Reversible inhibitor
- C. Competitive inhibitor
- D. Non-competitive inhibitor

295. **Malonic acid is an example of _____?**

- A. Irreversible inhibitor
- B. Reversible inhibitor
- C. Competitive inhibitor**
- D. Non-competitive inhibitor

296. **Magnifying power of electron microscope as compared to eye is _____?**

- A. 500 X
- B. 250000X**
- C. 500000X
- D. 250X

297. **Inhibitors which block the enzyme by forming weak bond are called _____?**

- A. Competitive inhibitors.
- B. Non-competitive inhibitors
- C. Irreversible inhibitors.
- D. Both a and b**

298. Which of the statement about cell membrane is not true?

A. It contains protein molecules embedded in lipid bilayer

B. It is a differentially permeable membrane.

C. It contains charged pores thus ions being charged particles cross cell membrane much easier than neutral particles.

D. It may get infolded to engulf solid or liquid material.

299. A substance which binds at the active site of the enzyme but does not result in the formation of the products is called_____?

A. Irreversible inhibitor

B. Reversible inhibitor

C. Competitive inhibitor

D. Non-competitive inhibitor

300. A chemical substance which can react (in place of substrate) with the enzyme but is not transformed into product/s and thus blocks the active site temporarily or permanently is called_____?

A. Co-enzyme

B. Blocker

C. Inhibitor

D. Cofactor

301. Optimal temperature of enzymes present in human body is_____?

A. 27°C

B. 37°C

C. 47°C

D. 30°C

302. Strengthening material of prokaryotic cell wall is_____?

A. Cellulose

B. Chitin

C. Inorganic salts silica waxes and ligin

D. Peptidoglycan or Murein.

303. Spherical or tubular membranes which separate the material present in endoplasmic reticulum from that of cytoplasmic material are called_____?

A. Cytosol

B. Cisternae

C. Polysomes

D. Cristae

304. Whone is not the function of endoplasmic reticulum?

A. Nerve impulse conduction

B. Transport of material

C. Mechanical support

D. Synthesis of conjugated molecules

305. 60S and 40S subunit combine to form _____ particle.

A. 100 S

B. 90 S

- C. 80 S
- D. 70 S

306. **Pancreas produces secretory granules that help in digestion. These granules after passing through endoplasmic reticulum are pinched off from _____ surface of Golgi apparatus:**

- A. Forming face
- B. Maturing face**
- C. Any of them
- D. None of these

307. **Which of the following statement is incorrect about Glyoxisomes?**

- A. They contain enzymes which help in conversion of fatty acids into carbohydrate
- B. They are abundant in soyabeans but absent in pea.
- C. They are present through out life of a plant and provide them whth energy through Glyoxylate cycle.**
- D. They are signle membranous organelles

308. **Ribosomes + m-RNA?**

- A. Polysome**
- B. Phlosome
- C. Polosome
- D. None of these

309. **Which of the following impart a red colour to Rose petals?**

- A. Chloroplast
- B. Chlorophyll
- C. Chromoplast**
- D. Leucoplast

310. **During digesting the phagocytosed food particles vesicles formed from fusion of phagocytic vacuole with the enzymes secreted by Golgi apparantus are called_____?**

- A. Lysosomes
- B. Primary lysosomes
- C. Secondary lysosomes**
- D. Food vacuole

311. **The bond formed when two or more atoms complete their electron shells by sharing electrons is called_____?**

- A. Ionic bond
- B. Hydrogen bond
- C. Covalent bond**
- D. Electrovalent bond

312. **Which one is the basic element of organic compounds _____?**

- A. Oxygen
- B. Hydrogen
- C. Sulphur
- D. Carbon**

313. Reactions in which simple substances are combined to form complex substances are called_____?

- A. Metabolic reactions
- B. Catabolic reactions
- C. Anabolic reactions**
- D. None of these

314. Which chemical component has the same % in bacterial as well as the mammalian cell_____?

- A. Water**
- B. Carbohydrate
- C. Proteins
- D. Lipids

315. The branch of biology which deals with the study of chemical compounds and the chemical processes in the fliving organisms is called_____?

- A. Chemistry
- B. Biochemistry**
- C. Molocular biology
- D. Both a and b

316. Which chemical component has the greatest contribution in the total mammalian cell weight_____?

- A. Proteins
- B. Carbohydrate
- C. Lipids
- D. Water**

317. Which one is an organic compound_____?

- A. CO₂
- B. Water
- C. Lipids**
- D. HCl

318. Carbon atom is_____?

- A. Monovalent
- B. Divalent
- C. Trivalent
- D. Tetravalent**

319. The property of water due to which it works as a temperature stabilizer and hence protect living organisms from sudden thermal changes is_____?

- A. High specific heat of vaporization
- B. High specific heat capacity**
- C. Its Dipole nature
- D. Its liquid state

320. The number of calories required to raise the temperature of 1g of water from 15 to 16°C is called_____?

- A. Specific Heat of Vaporization
- B. Specific Heat capacity**

- C. Caloric Heat
- D. Both A and C

321. The one which is present in all living things is _____?

- A. Cell nucleus
- B. DNA or RNA**
- C. Cell membrane
- D. Vacuole

322. The mechanism of stomatal movement is related to the branch of Biology called _____?

- A. Taxonomy
- B. Physiology**
- C. Morphology
- D. Anatomy

323. Which of the following substance is most favorable to form structural component of biological membranes _____?

- A. Hydrophilic Carbohydrates
- B. Hydrophobic fats.**
- C. Both a and b
- D. None of these

324. Maximum number of species of living things on earth are _____?

- A. Algae
- B. Fungi
- C. Insects**
- D. Protozoa

325. Which statement is true about an aqueous medium _____?

- A. Ionic as well as non ionic substance in aqueous media retains their identity
- B. Enzymes can not perform catalysis reaction in this medium
- C. Ions and molecules move randomly thus are in more favorable state to react with other molecules and ions**
- D. It is less favorable media for chemical reactions

326. Which of the following carbohydrate is tasteless _____?

- A. Monosaccharide
- B. Oligosaccharide
- C. Polysaccharide**
- D. None of these

327. Which of the statement is not true for compounds like glycoprotein and glycolipids _____?

- A. They are conjugated molecules of carbohydrates
- B. Both have role in the extra cellular matrix of animals and bacterial cell wall
- C. They are components of biological membranes.
- D. Both are produced and secreted by endoplasmic reticulum**

328. A complex substance which on hydrolysis yields polyhydroxy aldehyde or ketone subunits is called _____?

- A. Lipid
- B. Carbohydrate**
- C. Protein
- D. Carotein

329. **Specific heat of vaporization of water is _____?**

- A. 574 Kcal/kg**
- B. 674 Kcal/kg
- C. 774 Kcal/kg
- D. 874 Kcal/kg

330. **C_n(H₂O)_n is a general formula of _____?**

- A. Monosaccharides
- B. Oligosaccharides
- C. Ploysaccharides
- D. Carbohydrates**

331. **Which one are most complex sugar _____?**

- A. Monosaccharides
- B. Oligosaccharides
- C. Polysaccharides**
- D. None of these

332. **Variety among amino acids is produced due to _____?**

- A. NH₂ group
- B. COOH group
- C. R group**
- D. All of these

333. **Primary structure of proteins determines _____?**

- A. Number of polypeptide chains involved
- B. Bending of polypeptide chains
- C. Amino acid sequence**
- D. Coiling of polypeptide chains

334. **Ribose is an example of _____?**

- A. Trioses
- B. Tetroses
- C. Pentose**
- D. Hexose

335. **The one which is not a globular protein _____?**

- A. Anti Rh-antibody
- B. Enzyme
- C. Myosin**
- D. Haemoglobin

336. **Starch cellulose and glaycogen yield _____ on complete hydrolysis?**

- A. Maltose
- B. Sucrose
- C. Fructose
- D. Glucose**

337. **_____ % of Glucose normally present in our blood is?**

- A. 0.80%
- B. 0.08%**
- C. 1.80%
- D. 8%

338. Which of the following carbohydrate can not be hydrolysed_____?

- A. **Monosaccharide**
- B. Oligosaccharide
- C. Polysaccharide
- D. None of these

339. An enzyme which converts a dipeptide into separate amino acids is an example of_____?

- A. Decarboxylase
- B. Hydrolase**
- C. Oxidoreductase
- D. Transferase

340. Which one is soluble in hot water_____?

- A. Starch
- B. Glycogen
- C. Amylose**
- D. Amylopectin.

341. For the synthesis of 10g of glucose _____ of energy is required?

- A. 717.6 Kcal**
- B. 727 Kcal
- C. 737 Kcal
- D. 747 Kcal

342. Cellular digestion is associated with which organelle_____?

- A. Mitochondria
- B. Golgi bodies
- C. Plastids
- D. Lysosomes**

343. The covalent bond between two monosaccharide subunits is called_____?

- A. Phosphodiester bond
- B. Peptide bond
- C. Glycosidic Bond**
- D. Ionic bond

344. The unicellular organisms ingest large molecules into their cytoplasm from the external environment without previously digesting them. This process is called_____?

- A. Diffusion
- B. Osmosis
- C. Phagocytosis**
- D. Plasmolysis

345. Which of the following polysaccharide is called animal starch_____?

- A. Starch
- B. Glycogen**

- C. Cellulose
- D. None of these

346. Which one gives blue colour with iodine_____?

- A. **Starch**
- B. Glycogen
- C. Cellulose
- D. Polysaccharide

347. **Mycoplasmas have been included in bacteria because:**

- A. Do not have a cell wall
- B. Are heterotrophic
- C. **Lack membrane bounded organelles**
- D. Reproduce by binary fission

348. Which one is abundant in animals_____?

- A. Starch
- B. **Glycogen**
- C. Cellulose
- D. None of these

349. The one which can tolerate highest external osmotic pressure is _____?

- A. Algae
- B. Amoeba
- C. **Fungi**
- D. Bacteria

350. The cell wall of oomycetes is chiefly composed of_____?

- A. Chitin
- B. **Cellulose**
- C. Lignin
- D. Proteins

351. _____ is example of pure form of cellulose?

- A. Silk
- B. Wool
- C. **Cotton**
- D. Paper

352. A compound produced as a result of a chemical reaction of an alcohol with an acid in which water molecule is released is called_____?

- A. Monosaccharide
- B. Fatty acid
- C. **Neutral lipid**
- D. Nucleic acid

353. Lipids are soluble in which of the following_____?

- A. Water
- B. **Ether**
- C. All solvents
- D. Not in any of the above

354. **One celled green ptotists are included in_____?**

A. Monera

B. Algae

C. Slime molds

D. Plants

355. **Lipid molecules can store double amount of energy as compared to same amount of carbohydrate because of high number of_____?**

A. C-C bonds

B. C-H bonds

C. C-N bonds

D. C-O bonds

356. **Fatty acids containing 18 C atoms and a single double bond is_____?**

A. Saturated Fatty acid

B. Unsaturated fatty acid

C. Oleic Acid

D. Plamitic acid

357. **Pick the odd one out**

A. Psilophyton

B. Rhynia

C. Cooksonia

D. Psilotum

358. **Which of the following statement is incorrect⁵ for fats containing unsaturated fatty acid_____?**

A. They contain double bond

B. They are usually solid at room temperature

C. They are lighter than water

D. Their specific gravity is less than 1

359. **Which one the following fatty acid is more soluble in an organic solvent and has higher melting point_____?**

A. Acetic acid

B. Butyric acid

C. Palmitic acid

D. All have same

360. **Peptide bond is_____?**

A. **C-N link**

B. C- O lind

C. N-H link

D. C-H link

361. **Animals obtain carbohydrates mainly from_____?**

A. Glucose

B. Starch

C. Sucrose

D. Glycogen

362. **The one which is incorrect pair?**

- A. Sori – Indusium
- B. Arthrophytes – Jointed
- C. Dichotomous – Vernation**
- D. Lycopods – Ground pries

363. **Double fertilization occurs in_____?**

- A. All plants
- B. All seed plants
- C. Gymnosperms
- D. Angiosperms**

364. **Proteins comprise of _____% of the total dry weight of a cell?**

- A. 40%
- B. 50%**
- C. 60%
- D. 70%

365. **Adult birds normally possess only one functional_____?**

- A. Ureter
- B. Ovary**
- C. Lung
- D. Testis

366. **Chemical component/s less in eukaryotic cell as compared to prokaryotic cell is / are (i) lipids (ii) carbodhydrates (iii) proteins (iv) DNA_____?**

- A. i & ii
- B. iii & iv
- C. i ii iii & iv
- D. iv only**

367. **Following possess bilateral symmetry as larva and radial symmetry as adult:**

- A. Chordata
- B. Hydra
- C. Echinodermata**
- D. Hemichordata

368. **Proteins are polymers of_____?**

- A. Fatty acids
- B. Isoprenoid units
- C. Amino acids**
- D. Nucleotides

369. **Which of the following is the empirical formula of chlorophyll b?**

- A. C₅₅H₇₀O₅N₄Mg
- B. C₅₅H₇₂O₆N₄Mg
- C. C₅₅H₇₀O₆N₄Mg**
- D. C₅₅H₇₂O₅N₄Mg

370. **In Glycin an amino acid R group is replaced by_____?**

- A. H
- B. CH₃
- C. COOH
- D. C = O

371. How many moles of carbon dioxide are produced by complete oxidation of one mole of puruvic acid?

- A. 1
- B. 2
- C. 3**
- D. 6

372. The element in basic structure of proteins which differentiate them from carbohydrates_____?

- A. C
- B. S**
- C. H
- D. N

373. Hunger pangs usually begin _____ after the previous meal.

- A. 2-4 hours
- B. 6-8 hours
- C. 10-12 hours
- D. 12-24 hours**

374. Total number of amino acids discovered so far in cells and tissues are_____?

- A. 20
- B. 25
- C. 150
- D. 170**

375. The major constituent of blood plasma is_____?

- A. Protein
- B. NaCl
- C. Water**
- D. Cholesterol

376. An insulin molecule consists of _____ polypeptide chains?

- A. 1
- B. 2**
- C. 3
- D. 4

377. The windpipe or trachea lies?

- A. Dorsal to oesophagus
- B. Ventral to oesophagus
- C. Lateral to oesophagus
- D. None of these**

378. A bond formed by linkage between -OH of carboxyl group of one amino acid and H of amino group of another amino acid which releases water is called_____?

- A. Phosphodiester linkage
- B. Glycosidic bond
- C. Peptide bond**
- D. Hydrogen bond

379. Total number of amino acids in a hemoglobin molecule are_____?

- A. 554
- B. 564
- C. 574**
- D. 584

380. Which structure of protein gives information about number and sequence of amino acids in it_____?

- A. Primary structure**
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

381. Locomotory structures are not found in which of the following group?

- A. Amoebas
- B. Zooflagellates
- C. Ciliates
- D. Apicomplexans**

382. Hemoglobin molecule exhibits which structural organization of proteins_____?

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure**

383. Yeasts reproduce asexually by forming_____?

- A. Asci
- B. Basidia
- C. Both a & b
- D. Buds**

384. In an aqueous environment the most stable tertiary conformation is that in which _____ amino acids are buried inside the conformation?

- A. Hydrophobic**
- B. Hydrophilic
- C. Basic
- D. None of these

385. Which of the following statement is incorrect about RNA_____?

- A. The pentose sugar in it is Ribose.
- B. It contains Adenine Guanine Thymine and Cytosine**

- C. It is present in the nucleoli
- D. It may be folded back on itself to give double helical characteristics.

386. **Protein coats of viruses are synthesized in_____?**

- A. Lytic cycle**
- B. Lysogenic cycle
- C. Both a & b
- D. All of these

387. **Each turn of DNA contain _____ nitrogenous base pairs?**

- A. 5
- B. 10**
- C. 20
- D. 25

388. **Membranes of the grane are sites where?**

- A. Oxygen is stored
- B. Food is stored
- C. Sunlight is trapped**
- D. CO₂ is stored.

389. **Antibodies play important role against microorganisms and other pathogens to which type of proteins do they belong_____?**

- A. Globular proteins**
- B. Fibrous Proteins
- C. Both a and b
- D. None of these

390. **The simplest of oxygen producing photosynthetic organisms are_____?**

- A. Algae
- B. Bacteria
- C. Cyanobacteria**
- D. Chlamydomonas

391. **Amphoixus is a_____?**

- A. Chordate
- B. Protochordate
- C. Lower chordate
- D. All of these**

392. **Which structural organization is most common in globular proteins_____?**

- A. Primary
- B. Secondary
- C. Tertiary**
- D. Quaternary

393. **Which of the following does not show quaternary structure_____?**

- A. Haemoglobin
- B. Pepsin

- C. Fibrin**
- D. Insulin

394. Which of the following occurs in anaerobic respiration but not in aerobic respiration?

- A. Release of CO₂
- B. Reduction of NAD
- C. Formation of ATP
- D. Production of ethanol from acetaldehyde**

395. helical structure is kept by the formation of _____ bonds among amino acids molecules?

- A. H bonds**
- B. Disulphide Bond
- C. Ionic bond
- D. Hydrophobic bonds

396. Which of the following are triploblastic and acoelomate?

- A. Sponges
- B. Annelida
- C. Platyhelminthes**
- D. Aves

397. The food of hydra consists of _____?

- A. Humus
- B. Detritus
- C. Dead microscopic animals
- D. Small crustaceans**

398. Actin and myosin are the basic proteins involved in contractile machinery of our body to which type of proteins do they belong _____?

- A. Globular proteins
- B. Fibrous Proteins**
- C. Both a and b
- D. None of these

399. If a plasmolysed plant cell is placed in water the cell will _____?

- A. Burst
- B. Be Killed
- C. Be deplasmolysed**
- D. Show no change in it

400. ATP is a an important molecule for its major function _____?

- A. As an energy currency of the cell**
- B. As a coenzyme
- C. Both a and b
- D. None of these

401. The first organism of whose genome was completely discovered was _____?

A. Human beings

B. Influenza Virus

C. Hemophilus Influenza

D. None of these

402. **All of the following veins carry deoxygenated blood except**

A. Superior vena cava

B. Inferior vena cava

C. Hepatic vein

D. Pulmonary vein

403. **Animals fats are _____ at room temperature?**

A. Liquids

B. Gases

C. Solids

D. None of these

404. **Sebum produced from sebaceous glands in a mammal helps in _____?**

A. **Protection against micro organisms**

B. Temperature regulation

C. Excretion

D. All of these

405. **Which one is not found in fatty acids in acylglycerols _____?**

A. C-6

B. C-25

C. C-26

D. C-30

406. **The one that stores calcium is _____?**

A. Sarcolemma

B. Sarcoplasm

C. Sarcoplasmic reticulum

D. Cytosol

407. **Which one of the following is most likely to occur in an animal during winter?**

A. Activation of sweat glands

B. Dilation of skin blood vessels

C. Panting

D. Thermogenesis

408. **In esterification OH comes from _____?**

A. Organic acid

B. Alcohol

C. Water

D. Sugar

409. **How many carbon atoms are required to form a furan ring (ribofuranose) _____?**

A. 3

B. 4

- C. 5
- D. 2

410. **Sleep movements are a type of_____?**

- A. Turgor movements**
- B. Growth movements
- C. Tactic movements
- D. Paratonic movments

411. **Keto group is represented as_____?**

- A. CO**
- B. C-COO-C
- C. HCOH
- D. HOH

412. **The type of learning in which there is loss or decrease in response to repeated stimuli?**

- A. Imprinting
- B. Habituation**
- C. Latent learning
- D. Insight learning

413. **Glycolipids and glycoproteins have structural role in the _____ matrix of animal and bacterial cell?**

- A. Extracellular**
- B. Intracellular
- C. Both of these
- D. Glucose

414. **The cells present in testes and secrete testosterone are_____?**

- A. Sertoli cells
- B. Germinal cell
- C. Interstitial cells**
- D. Spermatocyte

415. **In $Cx(H_2O)_y$ x ranges from_____?**

- A. 1 to 3 thousand
- B. 3 to many thousand**
- C. 3 to 5 thousand
- D. 5 to 9 thousand

416. **Exposure to low temperature stimulates plants to flower. This is called_____?**

- A. Photoperiodism
- B. Thermotropism
- C. Vernalisation**
- D. Thermoregulation

417. **The basic structural unit of a chromosome is_____?**

- A. The centromere
- B. Nucleosome**

- C. Telomere
- D. Histone

418. In living organisms the lubricant which provides protection against damage resulting from friction is_____?

- A. Carbohydrates
- B. Fatty acid
- C. Water**
- D. Protein

419. The first stage of development in which a cavity appears is the_____?

- A. Neurula
- B. Morula
- C. Gastrula
- D. Balastula**

420. The trioses which are intermediate in respiration and photosynthesis are_____?

- A. Glyceraldehydes
- B. Dihydroxy acetone
- C. Both of these**
- D. None of these

421. The one which causes contraction of wall of the uterus during and after birth?

- A. ADH
- B. MSH
- C. Oxytocin**
- D. Progesterone

422. The monosaccharide found in some bacteria and occurring rarely are_____?

- A. Trioses
- B. Hexoses
- C. Tetroses**
- D. Pentoses

423. Which statement correctly describes the transcription of DNA?

- A. It produces another DNA molecule
- B. It produces mRNA**
- C. It is a semi conservation process
- D. It occurs at surface of ribosomes

424. Triglyceride is also known as_____?

- A. Natural lipid
- B. Neutral lipid**
- C. Neutral wax
- D. All of these

425. Carbon commonly combines with_____?

- A. H
- B. N & O

- C. P & S
- D. All of these**

426. The sequence of 3 bases on tRNA which is complementary to condon of mRNA is called_____?

- A. Gene
- B. Anticodon**
- C. Code
- D. Codon

427. The survival of an animal depends upon its ability to take some from its environment_____?

- A. Chemicals
- B. Hydrocarbons
- C. Organic molecules**
- D. Inorganic molecules

428. In humans the number of tetrads formed during mitosis is_____?

- A. 23
- B. 46
- C. 0**
- D. 4

429. Non polar organic molecules are _____ in water?

- A. Soluble
- B. Insoluble**
- C. Partially soluble
- D. Depends upon condition

430. Interconversion of carbohydrates proteins and lipids in living cells are an example of_____?

- A. Coordinated catabolic activities
- B. Coordinated anabolic activities
- C. Both A and B**
- D. Any one

431. Which one of the following is correct sequence of stages of prophase 1 of meiosis

- A. Diakinesis diplotene pachytene leptotene zygotene
- B. Zygotene leptotene pachytene diplotene diakinesis
- C. Leptotene zygotene pachytene diplotene diakinesis**
- D. Pachytene leptotene diplotene zygotene diakinesis

432. Which one of the following syndrome is the result of meiotic non dysjunction?

- A. Klinefelters syndrome
- B. Turners syndrome
- C. Downs syndrome
- D. All of these**

433. Which one of the following stage preceeds mitosis during cell cycle?

- A. G1 phase
- B. S phase
- C. G2 phase**
- D. M phase

434. **Most of the cellular secretions are _____ in nature?**

- A. Glycolipids
- B. Glycoproteins**
- C. Nucleohistones
- D. Proteins

435. **The red green color blindness is sex linked recessive condition in man. A father with normal vision and a color blind mother would expect to produce.**

- A. Color blind sons and daughter with normal vision**
- B. Sons with normal vision and color blind daughters
- C. Color blind sons colour blind daughters and daughters with normal vision
- D. Sons with normal vision colour blind daughter and daughters with normal vision

436. **Genes will not be found in gene pairs in the _____?**

- A. Muscle cell of worm
- B. Uterine cells of dog
- C. Sperm cells of frog**
- D. Guard cells of leaf

437. **rRNA constitutes _____% of the total cellular RNA?**

- A. 3-4 %
- B. 10-20%
- C. 80%**
- D. 90%

438. **A genome is _____?**

- A. Appearance of an individual
- B. Genetic basis of few characters
- C. Full set of genes of an individual**
- D. All of these

439. **Which conjugated molecules play important role in gene expression _____?**

- A. Glycolipids
- B. Glycoproteins
- C. Nucleohistones**
- D. None of these

440. **There are mainly _____ types of tRNA?**

- A. 170
- B. 20**
- C. 25
- D. 15

441. **The disease in which transmembrane carrier for the chloride ion is not produced is _____?**

- A. Hypercholesterolemia
- B. Cystic fibrosis**
- C. Severe combined immunodeficiency syndrome
- D. AIDS

442. The one which is a weed killer is _____?

- A. IAA
- B. NAA
- C. 2 4-D**
- D. GA

443. Five kingdom classification was proposed by _____?

- A. Carlous Linnaeus
- B. E-Chatton
- C. Robert Whittaker**
- D. Lynn Margulis and Karlene Schwartz

444. Organisms of which of the following kingdom have absorptive mode of nutrition _____?

- A. Prokaryotae
- B. Protoctista
- C. Animalia
- D. Fungi**

445. Which of the Euglenas (a Protoctista) character resembles that of animals _____?

- A. They can move
- B. They lack cell wall
- C. They do not have chlorophyll
- D. Both a and b**

446. The one which can break open a plasmid ring is _____?

- A. Ligase enzyme**
- B. Primase enzyme
- C. Restriction endonuclease
- D. ATP-ase enzyme

447. According to Lamarch evolution occurred as the result of _____?

- A. Natural selection
- B. The theory of recapitulation
- C. Inheritance of acquired characters**
- D. Over production

448. *Allium cepa* is the scientific name of _____?

- A. Onion**
- B. Amaltas
- C. Potato
- D. Tomato

449. Which of the following category is most general with its members least resembling with each other than the other categories _____?

- A. Species
- B. Genus
- C. Family
- D. Order**

450. The first eukarotes appeared about _____ years ago?

- A. 1.5 billion**
- B. 2.5 billion
- C. 3.5 billion
- D. 4.5 billion

451. Those organisms which can prepare their own food from simple inorganic material and can store energy are called _____?

- A. Autotrophs**
- B. Heterotrophs
- C. Procariotique
- D. Eu-cariotique

452. *Solanum tuberosum* is a scientific name of _____?

- A. Onion
- B. Amaltas
- C. Potato**
- D. Tomato

453. The structure which has been formed or modified from gill pouches in humans is _____?

- A. Gills
- B. Eustachian tube**
- C. Middle ear
- D. Lungs

454. Scurvy and beri – beri are caused by the deficiency of _____?

- A. Vitamin B and C
- B. Vitamin B and D
- C. Vitamin C and B**
- D. Vitamin A and B

455. The word which E-Chatton suggested for bacteria and blue green algae was _____?

- A. Autotrophs
- B. Heterotrophs
- C. Pro-cariotique**
- D. Eu-cariotique

456. The greatest diversity of animals in the lake is found in which one of the following zones?

- A. Limmnetic zone
- B. Littoral zone**
- C. Profundal zone
- D. None of these

457. **So far over _____ number of species of animals are known?**

- A. million
- B. 1 million
- C. 1**
- D. 2 million

458. **What is the characteristic feature of consumers?**

- A. Autotroph
- B. Heterotroph**
- C. Parasite
- D. Saprophyte

459. **Which of the following organisms have Chitin as a major structural component of their cell wall _____?**

- A. Prokaryotae
- B. Protoctista
- C. Plantae
- D. Fungi**

460. **All of the following are involved in nitrification except**

- A. Nitrobacter
- B. Nitrosomonas
- C. Rhizohium**
- D. All of these

461. **DNA or RNA of viruses is enclosed in _____ coat?**

- A. Protein**
- B. Carbohydrate
- C. Lipid
- D. DNA

462. **In most ecosystems the greatest amount of energy flows through the _____?**

- A. Secondary consumers
- B. Herbivores**
- C. Carnivores
- D. Decomposers

463. **Which of the following relationship is NOT an example of symbiosis?**

- A. Rhizobium and leguminous plants
- B. Marchantia sporophyte and gametophyte**
- C. Algae and fungi in a lichen
- D. Mycorrhizal fungi and tree roots

464. **Other name for Kingdom monera is _____?**

- A. Protoctista
- B. Prokaryotae**
- C. Archae
- D. None of these

465. **Animals with greatest number of similarities are grouped together in a/an_____?**

- A. Phylum
- B. Family
- C. Genus**
- D. Order

466. **Amoeba belongs to which kingdom_____?**

- A. Prokaryotae
- B. Plantae
- C. Monera
- D. Protista**

467. **Competition between species will be greatest if they attempt to occupy the same_____?**

- A. Ecosystem
- B. Community
- C. Habitat
- D. Niche**

468. **The word virus is derived from Latin word venome meaning_____?**

- A. Sweet fluid
- B. Sore Fluid
- C. Poisonous Fluid**
- D. Salty Fluid

469. **The one which is present in all aerobic species is _____?**

- A. **Cytochrome c**
- B. Cytochrome a
- C. Cytochrome a₃
- D. Cytochrome b

470. **The size of smallest virus is_____?**

- A. 10 nm
- B. 20 nm**
- C. 30 nm
- D. 40 nm

471. **What percentage of light reaching the earth is used in photosynthesis?**

- A. 25%
- B. 10%
- C. 5%
- D. 1%**

472. **Viruses are _____ smaller than bacteria?**

- A. 10 – 100 times
- B. 10 – 1000 times**
- C. 100 – 1000 times
- D. 1000 – 10000 times

473. **Kangaroo rat most probably would be found in_____?**

- A. Tropical rain forest
- B. Tundra
- C. Grass land
- D. Desert**

474. **Vaccine Was discovered by_____?**

- A. Louis Pasteur
- B. Edward Jennar**
- C. Ivanowski
- D. Stanely

475. **The name Bacteriophage was coined by_____?**

- A. Charles Chamberland
- B. Ivanowski
- C. Twort
- D. D Herelle**

476. **Protein subunits of capsid is called the capsomeres is a characteristic of each virus. Their number in Herpes virus is_____?**

- A. 100
- B. 152
- C. 162**
- D. 172

477. **Which of the statement about viruses is incorrect_____?**

- A. We cannot grow them in laboratories on artificial medias
- B. They are obligate intracellular parasite.
- C. They can synthesize their nucleic acid**
- D. They are resistant to most of antibiotic treatment

478. **A virion is_____?**

- A. Virus**
- B. Viral protein
- C. Viral lysozyme
- D. Viral gene

479. **For which of the following no preventive vaccine is available?**

- A. HAV
- B. HBV
- C. HCV**
- D. HEV

480. **Which one of the following specially result in chronic liver disease leading to hepatic failure:**

- A. HAV
- B. HBV
- C. HCV**
- D. HEV

481. **Protein coats the capsid of adenovirus is_____?**

- A. Surrounded by genome
- B. Surrounding genome**
- C. Surrounding the envelope
- D. Both B and C

482. **Mode of transmission of hepatitis A is _____?**

- A. Serum
- B. Faeco-oral**
- C. Sexual contact
- D. All of these

483. **Attachment of the bacteriophage with the receptor site on the bacterial cell wall involves _____?**

- A. Strong covalent bond between virion and the receptor site.
- B. Weak chemical union between them**
- C. Both A and B depending upon the phage
- D. None of these

484. **Which of the following statement about AIDS is incorrect:**

- A. is a host specific disease
- B. An infected mother cannot pass the virus to her baby.**
- C. Results in enlargement of lymph nodes
- D. Can be prevented.

485. **Bacteriophage that infect E. coli are called _____?**

- A. M type
- B. N type
- C. T type**
- D. S type

486. **Complex symptoms like severe pneumonia vascular tumor sudden weight loss swollen lymph nodes and immune deficiency are features of:**

- A. Polio
- B. Hepatitis
- C. AIDS**
- D. HIV

487. **An isolated virus is not considered living since it _____?**

- A. Separates in to two inner parts
- B. cannot metabolize**
- C. Rapidly loses its Genome chemical inert
- D. Is coated with an air tight shield

488. **Which of the following is the smallest known virus:**

- A. Hepatitis virus
- B. Retrovirus
- C. Polio virus**
- D. Pox virus

489. **The agent of controversial nature causing mad cow infection is _____?**

- A. Bacteria
- B. Cowpox Virus
- C. Adenovirus
- D. Prion**

490. Which of the following virus has special affiliations with the tumor production:

- A. Hepatitis virus
- B. Retrovirus**
- C. Polio virus
- D. Pox virus

491. Which of the following is incorrect for the lysogenic cycle in life cycle of bacteriophage_____?

- A. Phage when gets incorporated with the bacterial chromosome is called prophage
- B. The bacterium continues to live and reproduce normally
- C. Lysogenic bacteria may get infected by the related phage**
- D. The cycle may convert into lytic type resulting from environmental exposure

492. An enzyme which can convert single stranded RNA into double stranded viral DNA is called_____?

- A. DNAase
- B. RNAase
- C. DNA polymerase
- D. Reverse Transcriptase**

493. In lysogenic cycle the process of separation of phage DNA from the hosts chromosome and initiation of lytic cycle is called_____?

- A. Lysis
- B. Lysogeny
- C. Induction**
- D. Adsorption

494. Which of the following disease results in Paralysis due to viral attack on C.N.S:

- A. Measles
- B. Herpes simplex
- C. Hepatitis
- D. Polio**

495. In life cycle of bacteriophage which step occur after attachment of prophage with the receptor site of the bacterial cell wall_____?

- A. Absorption
- B. Multiplication
- C. Lysogeny
- D. Penetration**

496. Tobacco mosaic virus is_____?

- A. Spherical
- B. Rod Shaped**

- C. Tadpole like
- D. Hexagonal

497. In the Lysogenic cycle the DNA of the bacteriophage_____?

- A. Joins the bacterial chromosomes
- B. Attaches to the inner surface of the host membrane
- C. Is immediately degraded when enters the host
- D. Goes directly to host ribosomes for translation

498. Which of the following is an RNA virus:

- A. Influenza virus
- B. Herpes virus
- C. HBV
- D. Pox virus

499. Temperate phage may exist as_____?

- A. Prophage
- B. Capsid
- C. Viron
- D. Retrovirus

500. In the lytic cycle of bacteriophage the host DNA is_____?

- A. Replicated
- B. Turn off by the protein coat.
- C. Digested into its nucleotides
- D. Turned on by the removal of the protein coat

501. Pepsinogen is produced from_____?

- A. Mucous cells
- B. Parietal cells
- C. Zymogen cells
- D. All of these

502. Carbohydrates, fats & proteins are essential for_____?

- A. Metabolic processes of an organism
- B. Providing energy
- C. Formation of structures
- D. All of these